# Capstone project @data26

**Part 1 : SQL and Part 2 : PowerBI**

## 1. Reminder on the context

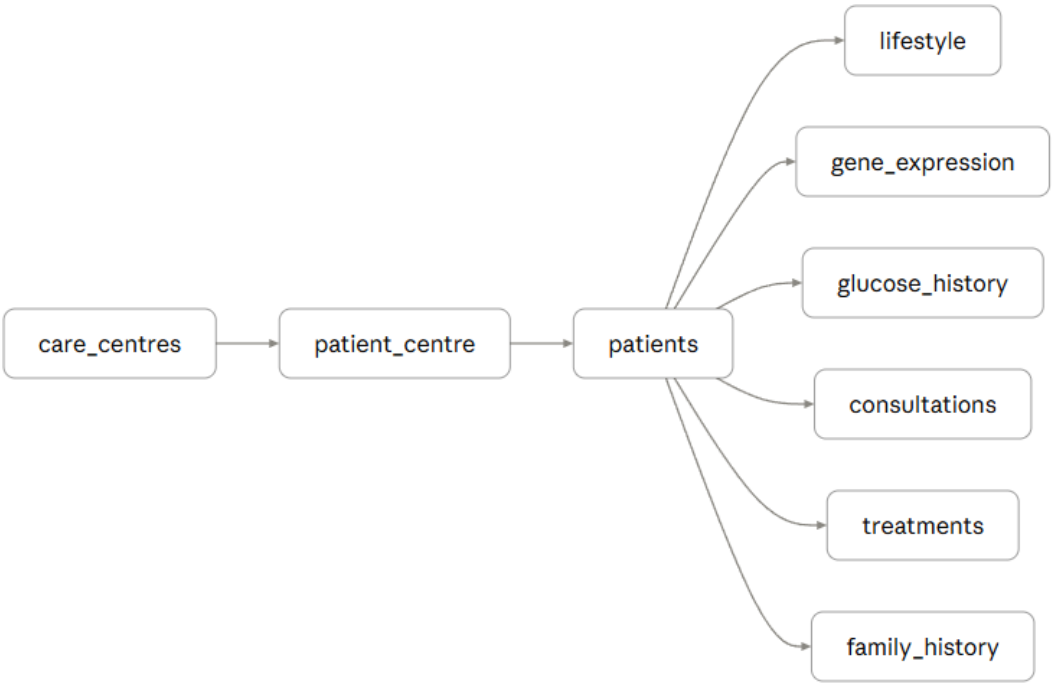
You will explore a fictitious cohort of 1,500 adult patients followed for diabetes in 24 Belgian care centres, first with SQL in DBeaver, then with Power BI, and finally compare the two. Each tool takes about 8 hours. Questions go step by step and most come with a hint. The data contain deliberate errors: spotting them is part of the job. Questions tagged **[X1]** to **[X10]** are answered in both tools and compared in Part C.

## 2. The dataset

Nine tables describe the patients. They are linked by the patient identifier (`P001` to `P1500`) and, for care centres, by the centre identifier (`C01` to `C24`).

| SQL table | Power BI workbook | Rows | One row per | Main columns |
|---|---|---|---|---|
| `patients` | Patients | 1,500 | patient | patient_id, age, sex, weight_kg, height_cm, blood_group, glucose_mg_dl, insulin_uiu_ml, corticosteroids |
| `lifestyle` | Hygiène vie | 1,500 | patient | patient_id, stress_level, activity_level, diet_quality, smoker, … |
| `gene_expression` | Expression génique | 1,500 | patient | patient_id + 19 genes |
| `patient_centre` | Patients centres | 1,500 | patient | patient_id, centre_id |
| `care_centres` | Centres soins | 24 | care centre | centre_id, centre_name, region, centre_type |

| SQL table | Power BI workbook | Rows | One row per | Main columns |
|---|---|---|---|---|
| `glucose_history` | Historique glycémique | 11,565 | measurement | patient_id, measure_date, glucose_mg_dl, weight_kg, blood_pressure, hba1c_pct |
| `consultations` | Consultations | 3,609 | consultation | patient_id, consult_date, consult_type, provider, reason |
| `treatments` | Traitements | 2,432 | treatment | patient_id, drug_name, start_date, end_date, dosage, drug_class |
| `family_history` | Antécédents familiaux | 1,662 | relative × condition | patient_id, relative, condition |



Every table is connected to `patients` through `patient_id`; care centres are reached through `patient_centre`.

### Glossary of French values

| Value in the data | Meaning |
|---|---|
| Oui / Non | Yes / No |
| Faible, Modéré(e), Élevé(e) | Low, Moderate, High |
| Peu équilibrée, Moyennement équilibrée, Équilibrée | Poor, Fair, Balanced (diet) |
| Diabète de type 2, Hypertension, Obésité, Maladie cardiovasculaire, Aucun antécédent connu | Type 2 diabetes, Hypertension, Obesity, Cardiovascular disease, No known history |
| Antidiabétiques, Corticoïdes, Antibiotiques, Antihistaminiques, Antidépresseurs, Aucun traitement | Antidiabetics, Corticosteroids, Antibiotics, Antihistamines, Antidepressants, No treatment |
| Médecine générale, Suivi diabétologique, Endocrinologie, Diététique, Prévention, Bilan | General practice, Diabetes follow-up, Endocrinology, Dietetics, Prevention, Check-up |

**Glucose categories** used in both tools: below 100 mg/dL = *Normal*, 100 to 125 = *Impaired*, 126 or more = *Diabetes range*.

## 3. Part A — SQL in DBeaver

Expected file format handed over to the HeLSci training team:

- Hand in one `.sql` script named `Diabetes_Cohort_<your name>.sql`.
- Put a comment such as `-- Q2.3` before each query,
- Write short answers to the *why* and *what do you conclude* questions as SQL comments.

| Part | Topic |
|---|---|
| 0 | Create the database |
| 1 | First queries: SELECT, WHERE, ORDER BY |
| 2 | Counting and grouping: COUNT, AVG, GROUP BY |
| 3 | Cleaning the data: UPDATE |
| 4 | Analysis on clean data |
| 5 | Combining tables: JOIN |

| Part | Topic |
|---|---|
| 6 | Cross-check with Power BI (Part C) |

**Part A0 — Create the database**

- Open DBeaver
- Navigate to “Postgres” => Databases on the left side
- Right click on “Databases” and click on “Create New database”
- Name it “Diabetes_capstone” and don’t change anything else
- Right click on the newly created database and click on “SQL Editor” => “New SQL script”
- A new tab opens on the main screen
- Navigate to the downloaded file “table_import.sql” from Digilearn and open it using Notepad/Bloc-note
- Select everything and copy paste its content into the “New script” window in DBeaver
- Type “ALT+X” on your keyboard to launch the script
- You should see a console showing the steps, then a table showing this :

Results 1 | Statistics 1

Grid | Text

| | table_name | n_rows | expected_rows |
|---|---|---|---|
| 1 | care_centres | 24 | 24 |
| 2 | patients | 1,500 | 1,500 |
| 3 | patient_centre | 1,500 | 1,500 |
| 4 | lifestyle | 1,500 | 1,500 |
| 5 | gene_expression | 1,500 | 1,500 |
| 6 | glucose_history | 11,565 | 11,565 |
| 7 | consultations | 3,609 | 3,609 |
| 8 | treatments | 2,432 | 2,432 |
| 9 | family_history | 1,662 | 1,662 |

If so, you are good to go with part A1.

If you break something later, delete the database, create it again and run `table_import.sql` again: it rebuilds everything from scratch.

## Part A1 — First queries

**Q1.1** Show the first 10 rows of `patients`. *Hint:* `SELECT * FROM … LIMIT 10;`

**Q1.2** List the patients older than 80 (`patient_id`, `age`, `sex`, `glucose_mg_dl`), oldest first. Look at the first row: what is wrong? *Hint:* `WHERE age > 80 ORDER BY age DESC`

**Q1.3** List the care centres located in Wallonia, sorted by name. *Hint:* `WHERE region LIKE 'Wallonie%'`

**Q1.4** List the different blood groups, without repeating them. *Hint:* `SELECT DISTINCT`

**Q1.5** List the patients whose glucose is 126 mg/dL or more (`patient_id`, `age`, `glucose_mg_dl`), highest glucose first. How many rows do you get?

**Q1.6** Show all treatments of patient `P335`, sorted by start date.

**Q1.7** Show all measurements of patient `P365` (date, glucose, weight, HbA1c), sorted by date. When does this patient's history start?

**Q1.8** List the patients whose sex is not recorded at all. *Hint:* an empty value is `NULL`, and you test it with `IS NULL`, not with `= NULL`.

## Part A2 — Counting and grouping

**Q2.1** How many rows are there in `patients`, in `glucose_history` and in `consultations`? (one query each)

**Q2.2** Return the minimum, maximum and average age (2 decimals), and the average glucose (1 decimal), of all patients. Is the average age reliable? **[X2]** *Hint:* `ROUND(AVG(age), 2)`

**Q2.3** Count the patients for each value of `sex`. How many different values do you see? How many patients are really women, men, or unknown? **[X1]** *Hint:* `SELECT sex, COUNT(*) FROM patients GROUP BY sex;`

**Q2.4** Same question for `corticosteroids`. Which values should not be there? **[X3]**

**Q2.5** Same question for `activity_level` in `lifestyle`. What is wrong?

**Q2.6** Count the measurements per year, with the number of different patients each year. What is unusual about 2020 to 2023? **[X4]** *Hint:* the year is `SUBSTR(measure_date, 1, 4)`; count patients with `COUNT(DISTINCT patient_id)`.

**Q2.7** For each `drug_class` in `treatments`, count the rows and the different patients, most frequent first. **[X7]**

**Q2.8** Duplicates. Find the rows that appear twice in `consultations`, then in `family_history`. *Hint:* group by **all** the columns and keep the groups with `HAVING COUNT(*) > 1`.

**Q2.9** Find the patients who have two measurements on the same date in `glucose_history`. *Hint:* `GROUP BY patient_id, measure_date HAVING COUNT(*) > 1`

**Q2.10** Which glucose values appear most often in `glucose_history`? Show the top 5. Is the first one realistic?

## Part A3 — Cleaning the data

You will now correct the errors found in Part 2. The **same rules** apply in Power BI, so that the two tools give the same results in Part C.

| Rule | Column | What to do |
|---|---|---|
| R1 | sex | `f`, `femme` (any case) → `F`; `m`, `h`, `homme` → `M`; everything else → `Unknown` |
| R2 | corticosteroids | `NA`, `na`, `Na` and empty → `Non` (after checking in Q3.2) |
| R3 | age | an impossible age (over 110) → `NULL` |
| R4 | activity_level | `faible` → `Faible` |
| R5 | consultations, family_history | remove the duplicate rows |
| — | everything else | keep and only report: measurements from 2020–2023, same-day measurements, values at exactly 58 mg/dL |

`UPDATE` changes the data for good. Always run a `SELECT` with the same `WHERE` first, to see which rows you are about to change. If you make a mistake, run `table_import.sql` again.

**Q3.1** Apply R1 with three `UPDATE` statements, then count the patients per sex again. **[X1]** *Hint:*

```
UPDATE patients SET sex = 'F' WHERE LOWER(sex) IN ('f', 'femme');
```

Write the one for men yourself. For the last one, remember that `NULL` is not caught by `NOT IN`: use `WHERE sex IS NULL OR sex NOT IN ('F', 'M')`.

**Q3.2** Before applying R2, check that none of these patients actually receives a corticosteroid: list the rows of `treatments` where `drug_class = 'Corticoïdes'` and whose patient has a strange `corticosteroids` value. How many rows do you get? Then apply R2 and count again. **[X3]** *Hint:* `WHERE patient_id IN (SELECT patient_id FROM patients WHERE …)`

**Q3.3** Apply R3, then compute the minimum, maximum and average age again. How much did the average change? **[X2]**

**Q3.4** Apply R4 and check `activity_level` again.

**Q3.5** Apply R5 by creating two clean copies, then count their rows:

```
CREATE TABLE consultations_clean AS SELECT DISTINCT * FROM consultations;
```

Do the same for `family_history` (→ `family_history_clean`). From now on, use the `_clean` tables.

## Part A4 — Analysis on clean data

**Q4.1** For each sex, give the number of patients, their percentage (1 decimal) and their average age. **[X1]** **[X2]** *Hint:* percentage = `ROUND(100.0 * COUNT(*) / 1500, 1)`

**Q4.2** Count the patients in each glucose category, with their average age. Complete this query:

```
SELECT CASE WHEN glucose_mg_dl < 100 THEN 'Normal'
            WHEN glucose_mg_dl < 126 THEN 'Impaired'
            ELSE 'Diabetes range' END AS glucose_category,
       COUNT(*) AS n_patients
FROM patients
GROUP BY glucose_category;
```

**Q4.3** Count the consultations per year in `consultations_clean`. By how many percent did they grow from 2024 to 2025 (calculate by hand)? Why should 2023 and 2026 not be compared with the other years? **[X5]**

**Q4.4** Count the consultations per type, most frequent first.

**Q4.5** How many measurements show a glucose of 126 mg/dL or more, and how many different patients do they concern? **[X10]**

**Q4.6** Give the number of measurements and the average HbA1c (2 decimals) per year. Does HbA1c change over time?

**Q4.7** How many treatments are still ongoing (no end date), not counting the rows `Aucun traitement`?

## Part A5 — Combining tables

**Q5.1** Join `patients` and `lifestyle` to give the number of patients and their average glucose for each stress level. What do you notice? **[X9]** *Hint:*

```
SELECT l.stress_level, COUNT(*) AS n_patients, ROUND(AVG(p.glucose_mg_dl), 1) AS
avg_glucose
FROM patients p
JOIN lifestyle l ON l.patient_id = p.patient_id
GROUP BY l.stress_level;
```

**Q5.2** Same question for `activity_level`, then for `smoker`. Which factor seems linked to glucose, and which does not?

**Q5.3** Count the patients per region. You need `patient_centre` and `care_centres`. **[X8]**

**Q5.4** Give the number of patients and the average glucose for each type of care centre. This time you need three tables.

**Q5.5** Compare the patients who have at least one relative with type 2 diabetes with the others: number of patients and average glucose for each group. **[X6]** *Hint:* two queries, one with `WHERE patient_id IN (SELECT patient_id FROM family_history_clean WHERE condition = 'Diabète de type 2')`, the other with `NOT IN`.

**Q5.6** For each antidiabetic drug, give the number of patients and their average glucose. Why is glucose so high for the three rare drugs?

**Q5.7** Critical thinking. Count the patients per stress level among those who received an antidepressant, then among the others. Would you expect this result? What does it say about the data?

**Q5.8** Find the last measurement date in `glucose_history` (`MAX`). Then list the treatments that start after that date. Is that possible?

### Bonus (optional, for those who finish early)

**B1** Compute the BMI (`weight_kg / ((height_cm / 100.0) * (height_cm / 100.0))`) and count the patients in each BMI category (below 18.5, 25, 30, and above), with their average glucose.

**B2** Find each patient's most recent measurement (a subquery with `MAX(measure_date)` joined back to `glucose_history`) and list the 10 highest HbA1c. How many rows does the join return in total, and why is it not 1,500?

**B3** Using `LAG(weight_kg) OVER (PARTITION BY patient_id ORDER BY measure_date)`, show patient P004's weight next to the previous weight. Where do you see a jump of more than 5 kg in a few days?

## Part B — Power BI

Expected file format handed over to the HeLSci training team:

- Hand in a `.pbix` file named `Diabetes_Cohort_<your name>.pbix`
- And the Part C answer sheet.

| Step | Topic |
|---|---|
| B1 | Import the data |
| B2 | Look at the raw data and write down what you see |
| B3 | Clean the data in Power Query |
| B4 | Relationships and measures |
| B5 | Report pages |
| C | Cross-check with SQL |

### B1 — Import the data

**B1.1** Import the nine files with *Get data › text/csv*.

**B1.2** Click **Transform data** (not *Load*) to open Power Query. Rename each query with a short English name: `Patients`, `Lifestyle`, `PatientCentre`, `CareCentres`, `GlucoseHistory`, `Consultations`, `Treatments`, `FamilyHistory`. Check that every column has the right type (numbers as decimal or whole numbers, dates as dates). Then *Close & Apply*.

### B2 — Look at the raw data

Power BI always shows the result of your Power Query steps. Once you clean the data in B3, the raw values are gone from your visuals. So **write down the raw values now**, in the *Raw* column of the Part C table.

**B2.1** Add a card with the number of patients (drag `ID patient` and choose *Count*). Add a card with the age. Power BI shows the **sum** of ages by default: change it to **Average**. Write the value down. **[X2]**

**B2.2** Add a bar chart of patients by `Sexe`. How many bars do you see? Compare with the number of values you found in SQL (Q2.3): why is it different? **[X1]**

**B2.3** Add a table with `Corticoïdes` and the count of patients. Write the values down. **[X3]**

**B2.4** Add a column chart of the number of measurements per year (use the *Year* level of `Date de mesure`). Write the values down. **[X4]**

**B2.5** Add cards with the number of rows of `Consultations` and of `FamilyHistory`.

You can delete this page once your notes are complete.

### B3 — Clean the data in Power Query

Apply the **same rules R1 to R5** as in SQL (Part A, Part 3), using the ribbon. Give every step a clear name (right-click › *Rename*), for example *Harmonised sex*.

**B3.1 — Sex (R1).** In `Patients`, select `Sexe` and use *Transform › Format › UPPERCASE*. Then *Transform › Replace values* several times: `FEMME` → `F`, `HOMME` → `M`, `H` → `M`, `NA` → `Unknown`, and `null` → `Unknown`. For short values like `H`, open *Advanced options* and tick **Match entire cell contents**: otherwise `HOMME` becomes `MOMME`. Check with the filter arrow of the column that only `F`, `M` and `Unknown` remain.

**B3.2 — Corticosteroids (R2).** Same method on `Corticoïdes`: uppercase, then `NA` → `NON` and `null` → `NON`, then *Format › Capitalize Each Word* to get back `Oui` / `Non`.

**B3.3 — Age (R3).** *Add column › Conditional column* named `Age`: if `Âge (ans)` is greater than 110 then `null`, otherwise the value of `Âge (ans)`. Set its type to whole number and remove the old column.

**B3.4 — Glucose category.** *Add column › Conditional column* named `Glucose category`: less than 100 → `Normal`, less than 126 → `Impaired`, otherwise `Diabetes range`.

**B3.5 — Activity level (R4).** In `Lifestyle`, select `Niveau d'activité` and use *Format › Capitalize Each Word*.

**B3.6 — Duplicates (R5).** In `Consultations`, select all columns (Ctrl+A) and use *Home › Remove rows › Remove duplicates*. Do the same in `FamilyHistory`.

**B3.7 — Region and centre type.** In `Patients`, use *Home › Merge queries* with `PatientCentre` on `ID patient` and expand `ID centre`. Merge again with `CareCentres` on `ID centre` and expand `Région` and `Type`.

**B3.8 — Lifestyle columns.** Merge `Patients` with `Lifestyle` on `ID patient` and expand `Niveau de stress`, `Niveau d'activité` and `Tabagisme`.

**B3.9 — Family history of diabetes.**

1. Right-click `FamilyHistory` › *Reference*. Name the new query `T2D relatives`.
2. Filter `Pathologie` on `Diabète de type 2`, keep only `ID patient`, then *Remove duplicates*.
3. In `Patients`, merge with `T2D relatives` (join kind *Left outer*) and expand `ID patient`.
4. Add a conditional column `Family T2D`: if the expanded column is `null` then `No`, otherwise `Yes`. Remove the expanded column.
5. Right-click `T2D relatives` and untick *Enable load*: it is only a helper.

Then *Close & Apply*. Check: `Patients` 1,500 rows, `Consultations` 3,608, `FamilyHistory` 1,652.

### B4 — Relationships and measures

**B4.1** In Power Query, right-click `Lifestyle`, `PatientCentre` and `CareCentres` and untick *Enable load*: their columns are now inside `Patients`.

**B4.2** In the *Model* view, check that `Patients` is linked to `GlucoseHistory`, `Consultations`, `Treatments` and `FamilyHistory` on `ID patient`: one-to-many (1 → *), single direction. Create the missing links by dragging `ID patient` from `Patients` onto the other table.

**B4.3** Create these measures (*Home › New measure*). The first two are given; the others follow the same pattern.

| Measure | Formula or description | Format |
|---|---|---|
| Patients | `Patients = COUNTROWS ( Patients )` | whole number |
| Avg Age | `Avg Age = AVERAGE ( Patients[Age] )` | 1 decimal |
| Avg Glucose | average of the patients' glucose | 1 decimal |
| Measurements | number of rows of `GlucoseHistory` | whole number |
| Consultations | number of rows of `Consultations` | whole number |
| Treated Patients | `DISTINCTCOUNT` of `ID patient` in `Treatments` | whole number |
| % Female | `DIVIDE ( CALCULATE ( [Patients], Patients[Sexe] = "F" ), [Patients] )` | % 1 decimal |
| High Glucose Measurements | `CALCULATE ( [Measurements], GlucoseHistory[Glycémie (mg/dL)] >= 126 )` **[X10]** | whole number |
| Ongoing Treatments | treatments without an end date, excluding `Aucun traitement` | whole number |

## B5 — Report pages

Use one theme for the whole report and put a title on each page.

**Page 1 — Overview**

- Cards: Patients, Avg Age, % Female, Avg Glucose.
- Bar chart: Patients by `Sexe`. **[X1]**
- Donut chart: Patients by `Glucose category`.
- Bar chart: Patients by `Région`, sorted from largest to smallest. **[X8]**
- Two slicers: `Sexe` and `Type` (centre type).

**Page 2 — Care and treatments**

- Bar chart: Treated Patients by `Catégorie Médicament`. **[X7]**
- Stacked column chart: Consultations by year, legend `Type de consultation`. **[X5]**
- Column chart: Measurements by year. **[X4]**

- Card: Ongoing Treatments.

**Page 3 — Risk factors**

- Bar charts of Avg Glucose by `Niveau de stress` **[X9]**, by `Niveau d'activité`, by `Tabagisme` and by `Family T2D` **[X6]**. Sort the levels in a logical order (low → high) using *Sort axis*.
- Table: `Type` (centre type), Patients, Avg Glucose.

**Page 4 — Data quality**

One card per problem found in the data, each with a short text box explaining it:

- Patients with unknown sex (card: Patients, visual filter `Sexe = Unknown`).
- Patients with a missing age (visual filter: `Age` is blank).
- Measurements before 2024 (visual filter on the year of `Date de mesure`).
- Measurements at exactly 58 mg/dL.
- Treatments starting after 28 June 2026, the last measurement date.

## B6 — Insight questions

Answer each question with the value and the visual you used.

1. Which region has the most patients, and which the fewest?
2. Which patients have the higher average glucose: those with a relative with type 2 diabetes, or the others? By how much?
3. How many consultations were there in 2024 and in 2025? Why should you not compare 2026 with 2025?
4. Which drug class concerns the most patients after antidiabetics?
5. Use the `Sexe` slicer: does the glucose category distribution look different for women and men?

## Bonus (optional)

- **Map:** show the patients per region on a map (ask your trainer for the dedicated exercise sheet).
- **Drill-through:** a page showing one patient's measurements over time, reached by right-clicking a patient in a table.
- **Date slicer:** a slicer on the year of consultation, and a card showing the consultations of the selected year.

## Part C — Cross-check SQL vs Power BI

Copy this table into your answer sheet and fill it in. *Raw* = before cleaning (SQL Part 2, Power BI B2); *Clean* = after applying rules R1 to R5.

| ID | Indicator | SQL | Power BI | Raw | Clean in SQL | Clean in Power BI | Same? If not, why? |
|---|---|---|---|---|---|---|---|
| X1 | Patients per sex | Q2.3, Q3.1 | B2.2, page 1 | | | | |
| X2 | Average age | Q2.2, Q3.3 | B2.1, Avg Age | | | | |
| X3 | Patients with corticosteroids Oui / Non | Q2.4, Q3.2 | B2.3, Patients table | | | | |
| X4 | Measurements per year | Q2.6 | B2.4, page 2 | | | | |
| X5 | Consultations in 2024 and 2025 | Q4.3 | page 2 | | | | |
| X6 | Average glucose with / without a T2D relative | Q5.5 | page 3 | | | | |
| X7 | Patients per drug class | Q2.7 | page 2 | | | | |
| X8 | Patients per region | Q5.3 | page 1 | | | | |
| X9 | Average glucose per stress level | Q5.1 | page 3 | | | | |
| X10 | Measure | Q4.5 | High | | | | |

| ID | Indicator | SQL | Power BI | Raw | Clean in SQL | Clean in Power BI | Same? If not, why? |
|---|---|---|---|---|---|---|---|
| | ments with glucose ≥ 126 mg/dL | | Glucose Measure ments | | | | |

**C1** When SQL and Power BI disagree, find out why: a rule applied differently, a step forgotten, a filter left on a visual, rounding. Correct the mistake and say in which tool it was.

**C2** Before cleaning, SQL and Power BI did not show the same number of values for `sex` (X1). Explain why.

**C3** Which cleaning rule changed the results the most? Which errors did you keep and only report (page 4), and why is it better not to delete them?

## How you will be assessed

| Criterion | SQL | Power BI |
|---|---|---|
| The queries or visuals give the right results | 60 % | 60 % |
| Clarity: comments, names, layout | 15 % | 20 % |
| Observations and critical thinking (the *why* questions) | 15 % | 10 % |
| Part C table | 10 % | 10 % |
| Bonus | up to +10 % | up to +10 % |